

HAMZA ELAHI 2023220

AI-311

Question 1:

Code:

```
PREFIX nobel: <http://data.nobelprize.org/terms/>

SELECT DISTINCT (REPLACE(STR(?cat), "^./category/", "")) AS
?categoryName)

WHERE {

    ?a a nobel:NobelPrize ;

        nobel:category ?cat .

}

ORDER BY ?categoryName
```

Output:



Table

≡ Response

6 results in 0.522 seconds

categoryName

- 1 http://data.nobelprize.org/terms/Chemistry
- 2 http://data.nobelprize.org/terms/Economic_Sciences
- 3 http://data.nobelprize.org/terms/Literature
- 4 http://data.nobelprize.org/terms/Peace
- 5 http://data.nobelprize.org/terms/Physics
- 6 http://data.nobelprize.org/terms/Physiology_or_Medicine

Showing 1 to 6 of 6 entries

Question 2:

Code:

```
PREFIX nobel: <http://data.nobelprize.org/terms/>
```

```
PREFIX foaf: <http://xmlns.com/foaf/0.1/>
```

```
SELECT DISTINCT ?personName
```

```
WHERE {
```

```
    ?chemPrize a nobel:NobelPrize ;
```

```
        nobel:category nobel:Chemistry ;
```

```
        nobel:laureate ?laureate .
```

```
    ?laureate foaf:name ?personName .
```

```
}
```

```
ORDER BY ?personName
```

Output:

Table		Response	193 results in 0.88 seconds
personName			
1	Aaron Ciechanover		
2	Aaron Klug		
3	Ada E. Yonath		
4	Adolf Butenandt		
5	Adolf Windaus		
6	Adolf von Baeyer		
7	Ahmed Zewail		
8	Akira Suzuki		
9	Akira Yoshino		
10	Alan Heeger		
11	Alan MacDiarmid		
12	Alfred Werner		
13	Archer J.P. Martin		
14	Arieh Warshel		
15	Arne Tiselius		
16	Arthur Harden		
17	Artturi Virtanen		
18	Avram Hershko		
19	Aziz Sancar		
20	Benjamin List		
21	Bernard L. Feringa		
22	Brian Kobilka		
23	Bruce Merrifield		
24	Carolyn Bertozzi		
25	Charles J. Pedersen		
26	Christian Anfinsen		
27	Dan Shechtman		
28	David Baker		
29	David W.C. MacMillan		
30	Demis Hassabis		
31	Derek Barton		
32	Donald J. Cram		
33	Dorothy Crowfoot Hodgkin		
34	Dudley R. Herschbach		
35	Eduard Buchner		
36	Edwin M. McMillan		
37	Ei-ichi Negishi		
38	Elias James Corey		
39	Emil Fischer		
40	Emmanuelle Charpentier		
41	Eric Betzig		
42	Ernest Rutherford		
43	Ernst Otto Fischer		
44	F. Sherwood Rowland		
45	Frances H. Arnold		
46	Francis W. Aston		
47	Frederick Sanger		
48	Frederick Soddy		
49	Fritz Haber		
50	Fritz Pregl		

Showing 1 to 50 of 193 entries

Question 3:

Code:

```
PREFIX nobel: <http://data.nobelprize.org/terms/>
```

```
PREFIX foaf: <http://xmlns.com/foaf/0.1/>
```

```
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
```

```
SELECT DISTINCT (COALESCE(?personFoafName, ?personLabel) AS  
?winnerName)
```

```
WHERE {
```

```
    ?awardNode nobel:laureate ?laureateNode ;
```

```
        rdfs:label ?awardLabel .
```

```
    FILTER(lang(?awardLabel) = "en")
```

```
    FILTER(regex(?awardLabel, "Chemistry", "i"))
```

```
    FILTER(regex(?awardLabel, "1911", "i"))
```

```
    OPTIONAL { ?laureateNode foaf:name ?personFoafName . }
```

```
    OPTIONAL { ?laureateNode rdfs:label ?personLabel .  
    FILTER(lang(?personLabel) = "en") }
```

```
}
```

```
ORDER BY ?winnerName
```

Output:



Table



Response

1 result in 1.086 seconds

winnerName

1 Marie Curie

Showing 1 to 1 of 1 entries

Question 4:

Code:

PREFIX nobel: <http://data.nobelprize.org/terms/>

PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

PREFIX foaf: <http://xmlns.com/foaf/0.1/>

PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>

SELECT DISTINCT ?winnerName ?yearExtracted

WHERE {

 ?awardNode nobel:laureate ?laureateRef ;

 rdfs:label ?awardText .

 FILTER(lang(?awardText) = "en")

 FILTER(regex(?awardText, "Chemistry", "i"))

 ?laureateRef foaf:name ?winnerName .

 BIND(xsd:integer(REPLACE(?awardText, "[^0-9]", "")) AS ?yearExtracted)

}

ORDER BY ?yearExtracted

Output:

Table		Response	195 results in 2.011 seconds	Simple view	Ellipse	Filter query results	Page size: 10		
winnerName		yearExtracted							
1	Jacobus H. van 't Hoff	"1901"^^xsd:integer							
2	Emil Fischer	"1902"^^xsd:integer							
3	Svante Arrhenius	"1903"^^xsd:integer							
4	Sir William Ramsay	"1904"^^xsd:integer							
5	Adolf von Baeyer	"1905"^^xsd:integer							
6	Henri Moissan	"1906"^^xsd:integer							
7	Eduard Buchner	"1907"^^xsd:integer							
8	Ernest Rutherford	"1908"^^xsd:integer							
9	Wilhelm Ostwald	"1909"^^xsd:integer							
10	Otto Wallach	"1910"^^xsd:integer							

Showing 1 to 10 of 195 entries

< 1 2 3 4 5 ... 20 >

Question 5:

Code:

PREFIX nobel: <<http://data.nobelprize.org/terms/>>

PREFIX foaf: <<http://xmlns.com/foaf/0.1/>>

PREFIX rdfs: <<http://www.w3.org/2000/01/rdf-schema#>>

SELECT DISTINCT ?femaleName

WHERE {

 ?laureateNode a nobel:Laureate ;

 foaf:gender ?sexValue ;

 foaf:name ?femaleName .

 FILTER(regex(?sexValue, "^female\$", "i"))

}

ORDER BY ?femaleName

Output:

Table		Response	64 results in 0.44 seconds	Simple view ☐ Ellipse ☒	Filter query results	Page size: 10	Download	Help
femaleName								
1	Ada E. Yonath							
2	Alice Munro							
3	Alva Myrdal							
4	Andrea Ghez							
5	Anne L'Huillier							
6	Annie Ernaux							
7	Aung San Suu Kyi							
8	Barbara McClintock							
9	Bertha von Suttner							
10	Betty Williams							

Showing 1 to 10 of 64 entries

< 1 2 3 4 5 6 7 >

Question 6:

Code:

```
PREFIX nobel: <http://data.nobelprize.org/terms/>
PREFIX foaf: <http://xmlns.com/foaf/0.1/>
PREFIX dbp: <http://dbpedia.org/property/>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>
SELECT DISTINCT ?fullName ?dob ?dod
WHERE {
    ?prizeNode nobel:year "2000"^^xsd:integer ;
        nobel:laureate ?recipient .
    ?recipient foaf:name ?fullName ;
        dbp:dateOfBirth ?dob .
    OPTIONAL { ?recipient dbp:dateOfDeath ?dod }
}
ORDER BY ?fullName
```

Output:

Table Response 13 results in 2.445 seconds Simple view Ellipse Filter query results Page size: 10			
	fullName	dob	dod
1	Alan Heeger	"1936-01-22"^^xsd:date	
2	Alan MacDiarmid	"1927-04-14"^^xsd:date	"2007-02-07"^^xsd:date
3	Arvid Carlsson	"1923-01-25"^^xsd:date	"2018-06-29"^^xsd:date
4	Daniel L. McFadden	"1937-07-29"^^xsd:date	
5	Eric Kandel	"1929-11-07"^^xsd:date	
6	Gao Xingjian	"1940-01-04"^^xsd:date	
7	Herbert Kroemer	"1928-08-25"^^xsd:date	"2024-03-08"^^xsd:date
8	Hideki Shirakawa	"1936-08-20"^^xsd:date	
9	Jack Kilby	"1923-11-08"^^xsd:date	"2005-06-20"^^xsd:date
10	James J. Heckman	"1944-04-19"^^xsd:date	

Showing 1 to 10 of 13 entries < 1 2 >

Question 7:

Code:

PREFIX foaf: <<http://xmlns.com/foaf/0.1/>>

PREFIX dbo: <<http://dbpedia.org/ontology/>>

PREFIX rdfs: <<http://www.w3.org/2000/01/rdf-schema#>>

PREFIX rdf: <<http://www.w3.org/1999/02/22-rdf-syntax-ns#>>

SELECT DISTINCT ?person ?personName ?country

WHERE {

{

 ?person rdf:type foaf:Person ;

 foaf:name ?personName ;

 dbo:birthPlace ?place .

 ?place rdf:type dbo:Country ;

 rdfs:label "Sweden"@en .

 BIND("Sweden"@en AS ?country)

}

UNION

```

{
  ?person rdf:type foaf:Person ;
    foaf:name ?personName ;
    dbo:birthPlace ?place .
  ?place rdf:type dbo:Country ;
    rdfs:label "Finland"@en .

  BIND("Finland"@en AS ?country)
}
}

ORDER BY ?personName

```

Output:

Table Response 34 results in 0.298 seconds				Simple view ☐ Ellipse ☒		Filter query results	Page size: 10		
person		personName	country						
1	<http://data.nobelprize.org/resource/laureate/305>	Allvar Gullstrand	"Sweden"@en						
2	<http://data.nobelprize.org/resource/laureate/543>	Alva Myrdal	"Sweden"@en						
3	<http://data.nobelprize.org/resource/laureate/208>	Arne Tiselius	"Sweden"@en						
4	<http://data.nobelprize.org/resource/laureate/203>	Artturi Virtanen	"Finland"@en						
5	<http://data.nobelprize.org/resource/laureate/722>	Arvid Carlsson	"Sweden"@en						
6	<http://data.nobelprize.org/resource/laureate/936>	Bengt Holmström	"Finland"@en						
7	<http://data.nobelprize.org/resource/laureate/426>	Bengt I. Samuelsson	"Sweden"@en						
8	<http://data.nobelprize.org/resource/laureate/689>	Bertil Ohlin	"Sweden"@en						
9	<http://data.nobelprize.org/resource/laureate/520>	Dag Hammarskjöld	"Sweden"@en						
10	<http://data.nobelprize.org/resource/laureate/604>	Erik Axel Karlfeldt	"Sweden"@en						

Showing 1 to 10 of 34 entries

< 1 2 3 4 >

Question 8:

Code:

PREFIX nobel: <http://data.nobelprize.org/terms/>

PREFIX foaf: <http://xmlns.com/foaf/0.1/>

PREFIX dbp: <http://dbpedia.org/property/>

PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>

SELECT (ROUND(AVG(?winnerAge)*100)/100 AS ?avgAge)

WHERE {

 ?personNode a foaf:Person ;

 dbp:dateOfBirth ?birthDate ;

 nobel:laureateAward ?chemAward .

 ?chemAward nobel:category

 <http://data.nobelprize.org/resource/category/Chemistry> ;

 nobel:year ?prizeYear .

```

    BIND(xsd:integer(SUBSTR(STR(?birthDate), 1, 4)) AS ?birthYr)

    BIND(xsd:integer(?prizeYear) AS ?awardYr)

    BIND(?awardYr - ?birthYr AS ?winnerAge)

}

```

Output:

Table		Response	1 result in 0.363 seconds
avgAge			
1	"59.29"	^^xsd:decimal	

Showing 1 to 1 of 1 entries

Question 10:

Code:

```

PREFIX nobel: <http://data.nobelprize.org/terms/>

PREFIX foaf: <http://xmlns.com/foaf/0.1/>

PREFIX dbp: <http://dbpedia.org/property/>

PREFIX dbo: <http://dbpedia.org/ontology/>

PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>


SELECT (STRAFTER(STR(?categoryNode), "/category/") AS ?categoryName)
       (ROUND(AVG(?ageAtAward)*100)/100 AS ?avgAge)
WHERE {
    ?laureateRef a foaf:Person ;
                nobel:laureateAward ?awardNode .

    ?awardNode nobel:category ?categoryNode ;
                nobel:year ?awardYr .

```

```

OPTIONAL { ?laureateRef dbp:dateOfBirth ?birthVal }
OPTIONAL { ?laureateRef dbo:birthDate ?birthVal }
OPTIONAL { ?laureateRef foaf:birthday ?birthVal }
FILTER(BOUND(?birthVal))
BIND(xsd:integer(SUBSTR(STR(?birthVal),1,4)) AS ?birthYr)
BIND(xsd:integer(?awardYr) AS ?awardYearInt)
BIND(?awardYearInt - ?birthYr AS ?ageAtAward)
}
GROUP BY ?categoryNode
ORDER BY ?categoryName

```

Output:

Table		Response	6 results in 0.668 seconds	Simple view ☐ Ellipse ☒		Filter query results	Page size: 10	Download	Help
	categoryName				avgAge				
1	Chemistry				"59.29"^^xsd:decimal				
2	Economic_Sciences				"67.02"^^xsd:decimal				
3	Literature				"64.88"^^xsd:decimal				
4	Peace				"60.64"^^xsd:decimal				
5	Physics				"57.57"^^xsd:decimal				
6	Physiology_or_Medicine				"58.82"^^xsd:decimal				

Showing 1 to 6 of 6 entries

< 1 >